



Investigation report

Rejected take-off and runway overrun, Tarlton Springs - ATSB final report

Drafted 30/09/2026

INVESTIGATION WORKFLOW SUITE

Contents

Investigation summary.....	3
What happened.....	3
What the investigation found.....	3
What has been done as a result	4
Safety message.....	4
The occurrence.....	5
Before the take-off.....	5
The take-off and rejected take-off	5
After the aircraft stopped.....	6
Context.....	8
Personnel information	8
Aircraft information	8
Meteorology and light conditions.....	8
Recorded data.....	9
Aerodrome information.....	9
Operational information	10
Safety analysis	12
External Influences	12
Organisational Influences.....	13
Risk Controls.....	15
Local Conditions	17
Individual Actions	19
Occurrence Events.....	20
Findings.....	23
Contributing factors	23
Other factors that increased risk	24
Other findings.....	24
Safety issues and actions	26
Documented wildlife plan required only above a size threshold the aerodrome does not meet	26
Perimeter fence repair deferred for three budget cycles	26
Evening service approved as a schedule change, route risk assessment not reopened.....	26
Runway inspection required only at listed aerodromes, and the listing criterion is a strike.....	27
Rejected take-off training covers aircraft system failures below V1 only	27
Cabin floor proximity emergency lighting partially unserviceable	28
Distribution	28

Investigation summary

What happened

On 18 June 2026 at 17:30 local time, Saltbush Air flight SB512, a 50-seat twin-turboprop registered VH-SBQ, began its take-off roll on runway 09 at Tarlton Springs Aerodrome for Sydney, carrying 44 passengers, 2 flight crew and 1 cabin crew member. It was 5 minutes after the end of civil twilight, with no moon and the aerodrome in full darkness. It was the flight crew's fifth sector of the day, and no runway inspection had been carried out since the morning.

Shortly after the decision speed (V1) of 112 kt, the first officer saw two kangaroos crossing the runway from the northern side. The pilot in command rejected the take-off at 118 kt, 6 kt above V1. One second later, the No. 2 propeller and right main gear struck the animals, leaving reverse thrust available on the No. 1 engine only. The aircraft left the sealed surface at approximately 21 kt and stopped upright on the extended centreline, 62 m into the 90 m runway end safety area (RESA) and 28 m short of the end of the prepared area.

No evacuation was commanded, and passengers disembarked by the forward airstair. There were no serious injuries, but one cabin crew member reported bruising sustained during disembarkation. Both kangaroos were killed. The aircraft was substantially damaged, including:

- three No. 2 propeller blades damaged beyond limits
- distorted right main gear doors
- both right main tyres deflated
- punctures to the lower right fuselage.

The aerodrome was closed for 4 hours 20 minutes.

What the investigation found

The overrun resulted from two factors acting together, neither sufficient alone:

- the stop began 6 kt and 3 seconds after V1, consuming the 67 m accelerate-stop margin
- the strike removed half the available reverse thrust, reducing deceleration to about 15% below the certified value.

The operator required a runway inspection before the first departure after last light, but only at aerodromes listed in the operations manual, and an aerodrome could be listed only after a reported wildlife strike. Tarlton Springs had never been listed, although its log recorded kangaroos on the sealed runway on 11 occasions, because those sightings were never passed to the operator. No inspection was requested, no document prompted the crew to consider one, and the aerodrome would have carried one out if asked.

The evening service, the operator's first operation into Tarlton Springs after last light, was approved as a schedule change only, without reopening a route risk assessment that restricted the aerodrome to daylight operations.

Repair of a failed section of the northern perimeter fence had been deferred for three budget cycles, and that breach was the likely route by which the animals reached the runway. Because the aerodrome fell below the size threshold in AS-139, it was not required to hold a documented wildlife hazard management plan. This likely contributed, although that link was less firmly established than the others.

The operator's rejected take-off training covered only aircraft system failures, with no crew trained for a runway obstruction. This likely contributed to the decision to reject above V1. However, the

pilot in command stated that he departed from the rule deliberately, so the training gap may equally be a fleet-wide matter rather than a cause of this rejection.

Two other matters increased risk but did not contribute:

- partially unserviceable floor proximity emergency lighting, undetectable by the pre-flight test
- a duty pattern that routinely placed the last sector late in a long day.

Whether runway edge lighting intensity affected detection of the animals could not be resolved. A tailwind and an engine power loss were ruled out. The crew's prompt braking, faster than the fleet average, reduced the severity of the overrun.

What has been done as a result

The operator's Head of Flying Operations suspended the evening Tarlton Springs service. No safety action has been recorded in response to the safety issues identified in this report.

Safety message

A defence triggered only by the outcome it exists to prevent offers no protection until that outcome occurs. Operators should ensure wildlife controls can be activated by sightings and aerodrome records, not only by strikes, and should reassess routes whenever a schedule change alters operating conditions. Aerodromes below regulatory thresholds hold wildlife knowledge that operators rarely see, and asking for it costs nothing.

The occurrence

Times in this section are Coordinated Universal Time (UTC) unless stated otherwise. The operator's duty record placed the occurrence at 17:30 local time.

Before the take-off

On 18 June 2026, Saltbush Air flight SB512, a 50-seat twin-turboprop registered VH-SBQ, was scheduled from Tarlton Springs Aerodrome to Sydney. It was the last scheduled sector of the day and the fifth of the flight crew's duty, added that morning when the rostered crew reported unfit.

The aerodrome's daily movement area inspection, at 21:35 on 17 June before the first scheduled arrival, recorded the movement area as clear. No further runway inspection was carried out or requested that day.

The doors closed at 07:16, 40 minutes behind schedule, with 44 passengers, 2 flight crew and 1 cabin crew member on board. The occurrence notification recorded that the aerodrome was in darkness. The tower was open and rostered to close at 18:00 local. [NOT ESTABLISHED: time of last light or end of civil twilight at Tarlton Springs on 18/06/2026]

The first officer was pilot flying. The take-off was planned at 20,800 kg with flap 10, on a dry runway 09 with a 5 kt headwind component, and a decision speed (V1) of 112 kt. At 07:29:10 the tower cleared SB512 for take-off on runway 09, passing the wind as 090° at 6 kt. The tower did not offer a runway inspection and the crew did not request one. No wildlife transmission was made on the tower channel in the preceding 90 minutes. The first officer had reviewed the flight plan, charts, the ERSA entry and the company aerodrome brief page, and reported that none mentioned animals.

The take-off and rejected take-off

The first officer called "Power set" at 07:30:02. The flight data recorder (FDR) showed both power levers advanced together and both engines at take-off torque, 98.4% and 98.5%, and the pilot in command confirmed "Power set, both engines good". The 80 kt cross-check was made at 07:30:15 with no caution or warning active. The recorded headwind component was 4 to 6 kt throughout the roll, with no tailwind component.

At 07:30:29 the pilot in command called "V one" at 112 kt indicated airspeed. Two seconds later, at 116 kt, the first officer called: "Roos - roos, roos, on the left, they're on the -". She had seen two kangaroos moving onto the runway from the northern side, crossing towards the south; the occurrence notification placed them past the 1,100 m point.

The first officer estimated the range at first sight at around 200 m, qualifying that it could have been less. Immediately afterwards she told the pilot in command: "They came out of the dark on the left of the strip, crossed the centreline. I had them for - I don't know, two seconds. Maybe less." He replied: "I never had them. I was heads-in on the engine page after V one." His air safety report, however, described looking up after the call and seeing two large animals already on the sealed surface. The available evidence did not reconcile the two accounts.

At 07:30:32 the pilot in command called "Stopping. I have control." The take-off was rejected at 118 kt, 6 kt above V1 and 3 seconds after it, with both power levers at flight idle within 0.6 seconds. Reconstruction from recorded groundspeed placed the aircraft 1,152 m along runway 09, leaving 695 m of sealed surface. Engine torques had remained within 0.1% of each other, with no engine exceedance, caution or warning recorded.

At 07:30:33 the aircraft struck the animals. The FDR recorded a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g, and the cockpit voice recorder (CVR) captured a sound consistent with a heavy impact, lasting about 0.4 seconds, followed by airframe rumble. The right main landing gear and the No. 2 propeller took the impact. The occurrence notification recorded that both animals were struck; the pilot in command's air safety report described striking at least one. The runway survey recovered two carcasses, at 1,188 m and 1,221 m from the runway 09 threshold, and animal tissue was recovered from two No. 2 propeller blade roots.

Maximum braking, at 2,940 psi, was established at 07:30:34, 2 seconds after the decision to reject, with anti-skid active on all four wheels; against the operator's recorded fleet mean of 3.2 seconds, this was at the fast end of the fleet distribution. The first officer remarked that they had "took one on the prop."

At 07:30:36 the pilot in command called "Reverse - number two's not coming. Max brakes, max brakes." Reverse was selected on the No. 1 engine only; No. 2 propeller RPM had been decaying since 07:30:33 and the No. 2 reverse interlock did not make. Between 07:30:34 and 07:30:44 mean deceleration was 2.6 m/s^2 , about 15% below the certified accelerate-stop value for the weight and runway condition, which assumed symmetric reverse.

At 07:30:47 the pilot in command said "We're not stopping on it." The first officer called "End's coming up" at 07:30:49.

At 07:30:51 the aircraft left the sealed surface at approximately 21 kt according to the occurrence notification; the FDR recorded 21 kt indicated airspeed and 20 kt groundspeed. In the graded gravel of the runway end safety area (RESA), longitudinal deceleration fell to 1.9 m/s^2 . The aircraft stopped at 07:30:57, upright on the extended centreline, 62 m into the 90 m RESA, with the nose wheel 28 m short of the end of the prepared safety area. Continuous tyre marks ran from 1,160 m to the runway end, and the nose wheel rut continued to the stopping point. The stopping distance from 118 kt was 757 m.

After the aircraft stopped

At 07:30:59 the pilot in command asked whether the first officer was all right. At 07:31:04 he set the parking brake and stated that there were no fire indications and "I'm not evacuating"; the first officer agreed. Both fuel levers were moved to cut-off at 07:31:26. At 07:31:34 the pilot in command instructed the cabin to remain seated and advised that there was no fire.

Replying to the tower at 07:31:48, the pilot in command reported that the aircraft had gone off the end of runway 09, with no injuries reported and no fire, and requested activation of the aerodrome emergency plan and a vehicle. The tower activated the plan at 07:32:10 and dispatched a vehicle. At 07:33:02 the pilot in command briefed the cabin for a slow disembarkation by the forward door, stating "no rush, no evacuation."

A sound consistent with the forward airstair operating was recorded at 07:34:40, and disembarkation continued until 07:38:48; the occurrence notification recorded it as 4 minutes 8 seconds after the aircraft stopped. No evacuation was commanded, no slides were deployed, and the main cabin lights remained on throughout. Aerodrome staff attended with a vehicle; there was no fire. The first officer later reported noticing unlit sections of the aisle floor proximity strip in the aft cabin during the disembarkation.

There were no serious injuries. One cabin crew member later reported bruising sustained during the disembarkation. Both kangaroos were killed. The aircraft sustained substantial damage:

- three No. 2 propeller blades damaged beyond repairable limits, including one blade tip separation

- distorted right main gear doors
- both right main tyres deflated through the fuse plugs, with brake energy calculated at 84% of the certified maximum
- abrasion and two punctures to the lower right fuselage skin.

The aerodrome operator closed the aerodrome for 4 hours 20 minutes pending inspection and removal of animal remains and propeller debris. The aircraft was quarantined at Tarlton Springs. The FDR and CVR were removed at 12:40 on 18 June and sent to the operator's flight data section. Both flight crew were stood down and tested for drugs and alcohol, with negative results. The operator's Head of Flying Operations suspended the evening Tarlton Springs service.

Context

Times in this section are local time (Australian Eastern Standard Time, UTC+10) unless stated otherwise.

Personnel information

Pilot in command

The pilot in command had 8,410 hours total, 2,140 hours on type, and had been in command on the operator's twin-turboprop fleet for 3 years and 1 month. His most recent cyclic simulator session (11 March 2026) and line check (2 December 2025) were satisfactory. He had no prior occurrence history with the operator.

He reported seeing kangaroos on the Tarlton Springs strip in the evening on several previous occasions and mentioning them to ground staff, but had not filed a report on any of them.

First officer

The first officer had 1,655 hours total, 405 hours on type. Her most recent cyclic simulator session, on 28 April 2026, was satisfactory in all items. She had flown two previous sectors into Tarlton Springs, both in daylight, on 9 January 2026 and 17 February 2026.

Duty

Both flight crew signed on at 06:30. SB512 was the fifth of five sectors and had been added that morning when the rostered crew reported unfit. The occurrence was 11 hours 00 minutes into a duty planned to finish at 18:35, a planned duty of 12 hours 05 minutes against a company maximum of 12 hours 30 minutes. Duty and rest in the preceding 7 days were within company limits with no variations applied. Both described themselves as tired but not struggling. The first officer had a break in Sydney mid-duty.

Aircraft information

VH-SBQ was a 50-seat twin-turboprop, the heaviest scheduled aeroplane type at Tarlton Springs at 22,900 kg. A post-occurrence borescope inspection of both engines found no pre-existing damage and no evidence of any power anomaly before the impact. The anti-skid system tested serviceable, and the right main gear structure was within limits.

Emergency lighting

After the first officer reported unlit sections of the aisle floor strip, the operator functionally checked the cabin emergency lighting on 24 June 2026. On battery power, two of the six floor proximity escape path segments on the left side aft of row 16 did not illuminate, due to a failed ballast common to both.

The type's pre-flight emergency lighting test neither illuminates nor monitors the floor proximity segments, so the defect could not have been detected without a dedicated functional test. The last such test was 11 months before the occurrence, within the 24-month maintenance interval. The same ballast is fitted across the type. [NOT ESTABLISHED: serviceability of floor proximity lighting on other aircraft of the type]

Meteorology and light conditions

...

METAR YTLS 180630Z 09007KT 9999 FEW040 09/M01 Q1024 RMK RF00.0/000.0

METAR YTLS 180700Z 09006KT 9999 FEW040 08/M01 Q1024 RMK RF00.0/000.0

METAR YTLS 180800Z 09005KT 9999 FEW040 07/M02 Q1024 RMK RF00.0/000.0

...

These gave a headwind component of 5 to 7 kt on runway 09. The crew used 5 kt for performance; the recorded wind at the aircraft gave 4 to 6 kt throughout the roll.

On 18 June 2026, sunset was at 16:58, civil twilight ended at 17:25, and moonrise was at 22:41, with the waning crescent moon 14% illuminated. The take-off roll began at about 17:30, with no natural ambient light on the strip and no town lighting within sight. At the same time of day on 23 June 2026, the investigator found that the unlit graded strip either side of the sealed surface could not be seen from the runway centreline.

Recorded data

The operator's flight data section read out the flight data recorder (FDR) and quick access recorder. The cockpit voice recorder (CVR) and tower audio were transcribed and verified by a second investigator. The operator's recorded fleet mean time from a decision to reject a take-off to maximum braking was 3.2 seconds.

Aerodrome information

General

Tarlton Springs is a certified aerodrome operated by Tarlton Shire Council, which handled 61,400 passenger movements in 2025. Runway 09/27 is sealed; runway 09 ASDA was 1,847 m, followed by a 90 m runway end safety area (RESA) of graded gravel, then ungraded terrain.

Aerodrome Manual 6.3 provided for a single daily movement area inspection before the first arrival, and additional inspections on request of the reporting officer or a pilot in command. The reporting officer confirmed an inspection would have been carried out if requested. Animals found during inspections were dispersed.

Wildlife hazard management requirements

Aerodrome Standard AS-139 section 10.4 states:

> 10.4.1 The operator of a certified aerodrome must take reasonable measures to reduce the risk to aircraft operations presented by wildlife on or in the vicinity of the aerodrome.

>

> 10.4.2 The operator of a certified aerodrome must prepare, implement and maintain a documented wildlife hazard management plan where the aerodrome is used by aeroplanes having a maximum take-off weight exceeding 27,000 kg, or serves more than 350,000 passenger movements in a calendar year.

>

> 10.4.3 Where subsection 10.4.2 does not apply, the measures taken under 10.4.1 need not be documented, but the aerodrome operator must be able to demonstrate what measures are in place.

Tarlton Springs met neither threshold and was not in breach. It held no wildlife hazard management plan, hazard assessment or dispersal programme, and had no documented wildlife measures beyond the daily inspection.

Perimeter fence

In April 2023, works register item WR-221 recorded a defect in the northern boundary fence between the 700 m and 1,300 m markers: four displaced panels, strained wire, and animals observed passing through. Replacement, estimated at \$41,000, was deferred as unfunded at the 2023, 2024 and 2025 budget rounds. Item WR-263, raised in February 2026, recorded further deterioration.

On 23 June 2026 the investigator found a continuous 140 m section with its bottom two wires separated from the posts, and a clear animal pad through it at the 1,050 m mark, on the same side as the carcasses recovered at 1,188 m and 1,221 m.

Wildlife sighting log

The aerodrome's paper wildlife sighting log contained 46 entries between January 2025 and June 2026:

- 41 recorded kangaroos on or next to the movement area
- 38 of those 41 were timed between one hour before and two hours after last light
- 11 recorded animals on the sealed surface of runway 09/27.

The log was not reported to any operator, and no airline had ever asked for it.

Runway edge lighting

A photometric check on 25 June 2026, 7 days after the occurrence, found four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal stage 3 intensity; the remainder were within tolerance. The maintenance programme checked lamp serviceability but not intensity.

Operational information

Route risk assessment and the evening service

The operator's route risk assessment RRA-041 issue 3, dated November 2022, assessed:

- runway length and accelerate-stop margin
- a single runway with no lighting redundancy
- no alternate within 200 nm at night
- limited engineering support
- summer heat and density altitude.

It carried the restriction "daylight operations only pending review of night lighting and diversion fuel policy".

Change Management Record CMR-2025-118 introduced an evening Sydney to Tarlton Springs to Sydney rotation, six days per week, from 14 July 2025, the operator's first scheduled operation there after last light. It assessed the change as "schedule change only" with "No new hazard identified", answered "No" to whether a route risk assessment review was required, and did not refer to the daylight-only restriction. The Head of Flying Operations approved it on 22 May 2025. RRA-041 was not reopened; no issue 4 exists. The Safety Department only received the approved document. The approver had since left the company and was not interviewed. The pilot in command stated that no briefing accompanied the evening service, which simply appeared on the roster.

Wildlife procedures

Operations Manual Part A section 8.14 stated:

> 8.14.2 At an aerodrome listed in Appendix 8A, the pilot in command shall request a runway inspection from the aerodrome operator prior to the first departure after last light, and shall not commence the take-off roll until the result of that inspection has been passed.

>

> 8.14.3 At all other aerodromes, no specific action is required. Crews should remain alert to the possibility of wildlife on the movement area.

Appendix 8A listed an aerodrome "where a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months", and was reviewed annually by the Safety Department, last on 3 March 2025. On 18 June 2026 it listed nine aerodromes; Tarlton Springs, where no company wildlife strike had been reported, had never been listed, although its own log recorded animals on the sealed runway on 11 occasions. Part A had no provision for crews to report wildlife sightings and no route to listing other than a reported strike.

None of the documents the first officer reviewed before the flight (flight plan, charts, ERSA entry and company aerodrome brief page) mentioned animals. The pilot in command stated that he had expected Tarlton Springs to be listed in Appendix 8A.

Rejected take-off procedures and training

Operations Manual Part B section 4.6.3 limited rejection between 80 kt and V1 to:

- fire or fire warning
- engine failure
- a configuration warning
- loss of directional control
- the pilot in command's judgement that the aircraft was unsafe or unable to fly.

Section 4.6.4 stated:

> At or above V1 the take-off shall be continued. The decision to continue is made before V1 is reached.

Section 4.6 contained no guidance on an obstruction seen on the runway at any speed.

Recurrent training syllabus TRG-SYL-07 module 4, completed by 214 crew in the 2025 to 2026 cycle, contained four scripted aircraft system scenarios:

- engine failure at 90 kt
- engine fire at 100 kt
- configuration warning at 85 kt
- engine failure at V1 plus 5 kt, with the take-off continued.

No crew had been presented with an external obstruction scenario in the five training cycles on file. Both flight crew stated independently that every stop or go scenario they had been given was an engine failure below V1.

Performance data

At 20,800 kg with flap 10, the scheduled accelerate-stop distance required was 1,780 m against the ASDA of 1,847 m, a margin of 67 m, assuming the stop began at or before V1 with symmetric reverse thrust.

Safety analysis

External Influences

Documented wildlife plan required only above a size threshold the aerodrome does not meet

AS-139 section 10.4 places two levels of obligation on the operator of a certified aerodrome. Every operator must, under 10.4.1, 'take reasonable measures to reduce the risk to aircraft operations presented by wildlife on or in the vicinity of the aerodrome'. Only an aerodrome 'used by aeroplanes having a maximum take-off weight exceeding 27,000 kg, or [that] serves more than 350,000 passenger movements in a calendar year' must, under 10.4.2, 'prepare, implement and maintain a documented wildlife hazard management plan'. Below that threshold, 10.4.3 provides that the measures 'need not be documented', although the operator 'must be able to demonstrate what measures are in place'.

Tarlton Springs fell below both limbs. Its heaviest scheduled aeroplane was the 22,900 kg twin-turboprop used on the Saltbush Air service, and it handled 61,400 passenger movements in 2025. The aerodrome's certification documents and the aerodrome operator's own compliance file confirmed this. The aerodrome held no wildlife hazard management plan, no hazard assessment and no dispersal programme. Apart from the single inspection before the first arrival at Aerodrome Manual 6.3, it held no documentation of any wildlife measure. None of this was a breach of the standard. The standard's text and the aerodrome's traffic figures were documentary and not in dispute. It is virtually certain that the threshold existed and that it left Tarlton Springs outside the requirement for a documented plan.

That the aerodrome complied is what makes this an external influence rather than a compliance matter. At an aerodrome of this size, the standard's own threshold allowed a wildlife programme that was undocumented and unaudited, and in practice unfunded.

The more difficult question was whether that threshold influenced the occurrence. Two facts about the aerodrome's own knowledge are central.

- **The sighting log.** The aerodrome's wildlife sighting log for January 2025 to June 2026 held 46 entries. Of these, 41 recorded kangaroos on or adjacent to the movement area. Of those 41, 38 were timed between one hour before and two hours after last light, and 11 recorded animals on the sealed surface of runway 09/27. The log was a paper register kept in the aerodrome office. It was not reported to any operator, and the aerodrome reporting officer confirmed that no airline had ever asked for it.
- **The fence.** The council's aerodrome works register had identified the failure of the northern perimeter fence in April 2023. The failed section lay between the 700 m and 1,300 m runway markers, and animals had been observed passing through it. The replacement was deferred as unfunded at the 2023, 2024 and 2025 budget rounds. A further entry in February 2026 recorded the section as deteriorating further. On 23 June 2026, a site inspection found a continuous 140 m section with the bottom two wires separated from the posts and a clear animal pad through it at the 1,050 m mark. The kangaroos in this occurrence came onto the runway from the northern side, past the 1,100 m point.

The aerodrome therefore already held both the knowledge of the hazard and knowledge of the defect in the physical barrier. The argument for influence is that a documented plan was the one mechanism that would have connected the sighting log to the operators using the aerodrome. It

was also the one mechanism that would have placed the failed fence on a funded programme, rather than leaving it on a works register where it could be deferred without consequence.

The argument against influence is also substantial, and the investigation gave it weight:

- Section 10.4.1 already obliged the aerodrome operator to take reasonable measures, whatever its size.
- A documented plan is not the only means of repairing a fence or passing a sighting log to an airline. Either could have happened without one.
- The council's three deferrals were budget decisions. There was no evidence that a documented plan would have attracted funding any more successfully. A plan is a document, whereas the fence repair, estimated at \$41,000, required money.

The investigation weighed these points against the pattern of the occurrence as a whole. A documented plan brings two things: an audit trail and an identified owner. Each of the other controls examined in this occurrence failed for want of exactly those two things. On that basis, the investigation concluded that the absence of a requirement for a documented plan likely contributed to the occurrence. This conclusion rests on inference about what a documented plan would have changed, not on direct evidence that it would have changed the outcome. The contrary argument, particularly on funding, was not disproved. This finding therefore carries less confidence than the other contributing factors in this report, and it should be read as such.

The factors downstream of this one are argued in their own subsections:

- The deferral of the perimeter fence repair across three budget cycles, found to have very likely contributed to the kangaroos entering the runway strip, is argued under Organisational Influences.
- The operator's arrangement requiring a runway inspection only at listed aerodromes, with a wildlife strike as the listing criterion, found to have very likely contributed to the absence of an inspection before the departure, is argued under Risk Controls.

Organisational Influences

Perimeter fence repair deferred for three budget cycles

The aerodrome operator's works register documented the condition of the northern perimeter fence well before the occurrence. Item WR-221, raised in 04/2023, described the fence on the northern boundary between the 700 m and 1,300 m runway markers:

- four panels were displaced and the wire was strained
- animals had been seen passing through
- replacement was recommended at an estimated cost of \$41,000.

The item was deferred as unfunded at the 2023, 2024 and 2025 budget rounds. Each time, it was carried forward for review at the next cycle. Item WR-263, raised in 02/2026, recorded that the same section had deteriorated further.

The site inspection on 23/06/2026 found a continuous 140 m section of the northern fence with the bottom two wires separated from the posts. A clear animal pad ran through the fence at the 1,050 m mark.

It was virtually certain that the repair had been deferred across three budget cycles. The deferrals are set out in the aerodrome operator's own register, in its own words, and the condition of the fence was confirmed on site. None of this depended on reconstruction.

It was very likely that the deferral contributed to the kangaroos reaching the runway. A fence in the condition found does not exclude kangaroos. The animal pad at the 1,050 m mark was on the

northern side of the runway. The first officer saw the animals moving onto the runway from that side, past the 1,100 m point. The two carcasses were recovered at 1,188 m and 1,221 m from the runway 09 threshold. The breach was therefore directly upstream of the strike position, and it was the physical route by which the animals reached the movement area.

The aerodrome's wildlife sighting log for 01/2025 to 06/2026 shows the scale of the exposure the fence was meant to control:

- Of the log's 46 entries, 41 recorded kangaroos on or adjacent to the movement area.
- Of those 41, 38 fell between one hour before and two hours after last light.
- Eleven entries recorded animals on the sealed surface of runway 09/27.

The log was a paper register held in the aerodrome office. It was not reported to any operator, and the aerodrome reporting officer confirmed that no airline had ever asked for it.

The aerodrome operator therefore knew about, and had documented, both the hazard and the breach through which it reached the runway. That knowledge did not lead to the fence being repaired. Nor did it reach the airline operating into the aerodrome after last light.

The regulatory framework under which the aerodrome operator was not required to document its wildlife measures is discussed under External Influences. The timing of the departure relative to kangaroo movement is discussed under Local Conditions. The animals' movement onto the sealed surface is discussed under Occurrence Events.

Evening service approved as a schedule change, route risk assessment not reopened

The operator's current route risk assessment for Tarlton Springs is RRA-041 issue 3 (11/2022). It assessed:

- runway length and accelerate-stop margin
- a single runway with no lighting redundancy
- no alternate within 200 nm at night
- limited engineering support
- heat and density altitude in summer.

It imposed the operating restriction 'daylight operations only pending review of night lighting and diversion fuel policy'. No later issue exists.

Change Management Record CMR-2025-118 was approved by the Head of Flying Operations on 22/05/2025. It covered the 'addition of one evening rotation, Sydney to Tarlton Springs to Sydney, six days per week, from 14/07/2025'. That rotation was the first scheduled company operation into Tarlton Springs after last light.

The record assessed the change as a 'schedule change only'. It listed the aerodrome, aircraft type, crew qualifications and engineering support arrangements as existing, and stated 'No new hazard identified'. It answered 'No' to whether a route risk assessment review was required. The daylight-only restriction in RRA-041 is not referenced anywhere in CMR-2025-118, and RRA-041 was not reopened. The Safety Department's records show no involvement in the change beyond receipt of the approved record.

It was extremely likely that the evening service was approved in this way. Both documents are current controlled copies. The approver has since left the company and could not be interviewed, so the reasoning behind the assessment could not be confirmed directly.

The approval process tested whether the elements of the change were new. It did not test whether the existing assessment still held for the proposed operation. Each element listed in the change

record did already exist. Operation after last light did not, and it was the one condition the route risk assessment had expressly excluded.

It was likely that this contributed to the occurrence. A review of RRA-041 against a night operation would have considered two further matters in the same pass:

- the aerodrome's exposure to wildlife after dark
- the aerodrome's absence from Appendix 8A of the Operations Manual Part A.

Against this, the restriction named only night lighting and diversion fuel, and a review might have been confined to those two matters. RRA-041 issue 3 did not list wildlife among the hazards it assessed. For that reason, the contribution was assessed as likely rather than more strongly.

The Appendix 8A listing criterion determined whether a runway inspection was required at Tarlton Springs. That criterion is discussed under Risk Controls.

Risk Controls

Runway inspection required only at listed aerodromes, and the listing criterion is a strike

The operator had a procedure designed for this situation. Operations Manual Part A 8.14.2 required that, at an aerodrome listed in Appendix 8A, the pilot in command 'shall request a runway inspection from the aerodrome operator prior to the first departure after last light, and shall not commence the take-off roll until the result of that inspection has been passed.' At every other aerodrome, 8.14.3 stated that 'no specific action is required.'

The requirement therefore depended entirely on which aerodromes were listed. Appendix 8A listed an aerodrome where 'a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months.' Nine aerodromes were listed. Tarlton Springs was not listed and never had been. The list was last reviewed on 03/03/2025, four months before the evening service began.

Nothing in Part A allowed an aerodrome to be listed on any other evidence, and nothing asked a crew to report a sighting. The only way onto the list was a reported strike. The aerodrome's own wildlife sighting log for 01/2025 to 06/2026 showed that evidence of the hazard existed:

- 41 of its 46 entries recorded kangaroos on or adjacent to the movement area
- 38 of those 41 were timed between one hour before and two hours after last light
- 11 recorded animals on the sealed surface of runway 09/27.

The log was a paper register held in the aerodrome office. It was not reported to any operator, and the aerodrome reporting officer confirmed that no airline had ever asked for it. None of those entries was a company strike, so none of them could place Tarlton Springs on the list.

The effect was that the control could only be activated by the kind of event it existed to prevent. Because Tarlton Springs had no reported company strike, it was not listed. Because it was not listed, no inspection was required before the first departure after last light. The control was sound in its design, but its trigger meant it was not active at this aerodrome on the night. It was virtually certain that this limitation existed, since both the manual text and the appendix listing are documentary.

Had the requirement applied, an inspection would have been requested, and the take-off roll could not have started until the result was passed. The aerodrome's staff disperse animals when they find them. There is one qualification: an inspection completed some minutes before departure does not guarantee a clear runway at the start of the roll. With that qualification, it was very likely that the limitation contributed to no inspection being requested before the departure. That crew-level outcome, and its link to the two kangaroos entering the runway strip and moving onto the sealed

surface, are discussed with the individual actions. The finding that the documented wildlife plan was required only above a size threshold the aerodrome did not meet, and the approval of the evening service as a schedule change without reopening the route risk assessment, both bear on how this gap persisted. They are discussed with the external and organisational influences.

Rejected take-off training covers aircraft system failures below V1 only

Module 4 of recurrent training syllabus TRG-SYL-07 required each crew to 'demonstrate a correctly executed rejected take-off for an engine failure, an engine fire and a configuration warning, each initiated below V1, and... the decision to continue for a failure recognised at or after V1.' Its four scripted scenarios were:

- engine failure at 90 kt
- engine fire at 100 kt
- configuration warning at 85 kt
- engine failure at V1 plus 5 kt, continued.

Each was an aircraft system event. In the five training cycles held on file, no crew had been presented with an external obstruction scenario. In the 2025 to 2026 cycle alone, 214 crew completed the module.

Operations Manual Part B 4.6 was consistent with the syllabus. Section 4.6.4 stated that 'At or above V1 the take-off shall be continued.' The section contained no guidance on an obstruction seen on the runway at any speed, and no guidance for a case where continuing is itself hazardous. The pilot in command and the first officer each stated, independently, that every stop-or-go scenario they had been given involved an engine and arose below V1. The documents bear this out, and it was extremely likely that the training gap existed. The small remaining doubt reflects that scenario records older than five cycles were not available.

Whether this gap influenced the decision to reject 6 kt above V1 was argued both ways.

- **For:** the captain had about two seconds to decide, with no prior framework. The decision on the night was the first time he had considered the question at all, and a decision made unrehearsed in that time differs from one made against a briefed and practised frame.
- **Against:** the captain stated that he knew the rule and departed from it deliberately, not through unawareness. On that reading, obstruction training would not have changed what he knew. He rejected because he judged the alternative to be worse, and the training gap is a matter for the fleet rather than a cause of this reject.

The investigation accepted that the training gap likely contributed to the decision to reject above V1. Both readings are retained. The training gap helps explain the conditions under which the decision was made; it does not stand in for the captain's own judgement. The rejected take-off itself is discussed with the individual actions.

Cabin floor proximity emergency lighting partially unserviceable

The first officer noticed sections of the aisle floor strip unlit while walking back through the cabin during the disembarkation. A functional check on 24/06/2026, with the system armed and on battery power, found the following:

- two of the six floor proximity escape path segments, on the left side aft of row 16, did not illuminate
- the fault was traced to a failed ballast common to both segments.

It was virtually certain that the defect existed.

It was unlikely that the defect influenced the occurrence or its outcome. No evacuation was commanded, the main cabin lights remained on, and passengers left unhurriedly by the forward airstair within 4 minutes 8 seconds of the aircraft stopping.

The defect remains an important safety matter because it could not be detected in normal operations. The pre-flight emergency lighting test on this type illuminates the ceiling emergency lights and the exit signs. It neither illuminates nor monitors the floor segments, so no check available to the crew or a line engineer would have revealed the fault. The last dedicated functional test was 11 months before the occurrence, within the maintenance programme's 24-month interval.

In different circumstances, such as a fire indication, smoke, or a night evacuation over the wing, two dark segments in the aft cabin could make the difference between an orderly exit and a blocked one. The same ballast is fitted across the type. Because the monitoring interval cannot detect this failure, the condition of the rest of the fleet is unknown. It was very likely that this latent defect increased risk.

Local Conditions

Departure in the hour after last light, the peak of kangaroo movement

The take-off roll began at 17:30 local time. That was 32 minutes after sunset and 5 minutes after the end of civil twilight. Moonrise was not until 22:41, and the moon was a waning crescent, 14% illuminated. There was no natural ambient light on the runway strip and no town lighting within sight.

The investigator visited the site at the same time of day on 23/06/2026. The visit confirmed that the unlit graded strip either side of the sealed surface cannot be seen from the runway centreline. The aerodrome's wildlife sighting log for 01/2025 to 06/2026 held 41 entries recording kangaroos on or next to the movement area. Of these:

- 38 were timed between one hour before and two hours after last light
- 11 recorded animals on the sealed surface of runway 09/27.

The log is a paper register, and its completeness could not be verified. That limits how much weight the exact distribution can bear. However, the astronomical data are documentary, and the log's pattern is pronounced. It is extremely likely that the departure took place in full darkness, inside the period in which the aerodrome's own records show almost all of its kangaroo movement.

The first officer's account after the stop, recorded on the CVR, describes how the light condition shaped what the crew could see:

> 'They came out of the dark on the left of the strip, crossed the centreline. I had them for - I don't know, two seconds. Maybe less. There was nothing to see and then there were two of them.'

The animals were not seen until they were on the lit surface. That follows directly from the light condition, and the timing coincides with the recorded movement peak. It is therefore likely that departing in this period contributed to two kangaroos being on the sealed surface, unseen, until the aircraft was past V1.

On its own, a departure after last light is unremarkable. It mattered here because no runway inspection was requested before this first departure after last light. That omission is discussed in the Risk Controls subsection. The approval of the evening service as a schedule change, without the route risk assessment being reopened, is discussed under Organisational Influences.

Runway edge lighting intensity reduced the crew's ability to detect the animals

The investigation examined whether the runway edge lighting reduced the crew's ability to see the animals. This could not be resolved. The evidence was about as likely as not to support the lighting deviation having been present on the night, and no conclusion is drawn from it either way.

Several things supported the possibility:

- **The captain's account.** In interview, the captain described the lighting as brighter than he would want. He said it formed a corridor of light with everything outside it black, so that anything off the seal could not be seen coming.
- **The photometric check.** A check at the investigator's request on 25/06/2026 found four edge lights on the northern side, between the 900 m and 1,200 m markers, at between 118% and 131% of nominal intensity for stage 3. This was the section the animals crossed. The remaining lights were within tolerance.
- **The maintenance programme.** The aerodrome operator's maintenance programme checked lamp serviceability but did not measure intensity. A deviation of this kind would therefore not have been detected in routine maintenance.

Several things weighed against it:

- **The first officer's account.** The first officer was the crew member who actually saw the animals. She described the lighting as normal and said she saw them clearly once they were on the sealed surface, at around 200 m. She also acknowledged that, with only two previous daylight sectors into Tarlton Springs, she was not well placed to judge whether the lighting was unusual.
- **The captain's position after V1.** The captain's non-detection has an explanation that does not involve the lighting. On the CVR he stated: 'I never had them. I was heads-in on the engine page after V one.' That is where the procedure placed his attention.
- **The timing of the measurement.** There is no measurement of the lights on the night of the occurrence, only the check seven days later.
- **No established effect.** Nothing establishes that an edge light at 131% of nominal affects the detection of an object on the runway.

Both crew accounts appear to be honest, and they do not agree. The one measurement available shows a deviation in the relevant section, but it does not show an effect on detection. The matter could only be settled with two things:

- evidence of the lighting output on the night itself
- evidence of how edge-light intensity of this order affects the detection of animals on the runway surface.

Neither was available. The matter therefore remains unresolved and is not carried into the findings.

Take-off commenced with a tailwind above limits

The observations at 06:30, 07:00 and 08:00 UTC gave a headwind component of 5 to 7 kt, and the recorded aircraft data gave a headwind of 4 to 6 kt throughout the roll. Because both sources agree, a tailwind was very unlikely and was ruled out.

Fifth sector, 11 hours into duty, 40 minutes behind schedule

The crew signed on at 06:30 local time. The fifth sector was added that morning when the rostered crew reported unfit. The occurrence was 11 hours 00 minutes into a planned 12 hour 05 minute duty, against a limit of 12 hours 30 minutes, and the aircraft departed 40 minutes behind schedule. Both crew described themselves as tired but not struggling. It is virtually certain that this exposure existed.

A late fifth sector at 11 hours into a duty is within the range where degraded decision-making would be expected. However, each mechanism through which fatigue could have acted in this occurrence was contradicted by evidence:

- **Response on the roll.** The power levers reached flight idle within 0.6 seconds of the decision, and maximum braking was established 2 seconds after it. This was at the fast end of the operator's fleet distribution, which has a mean of 3.2 seconds.
- **Errors of omission.** No error of omission was identified on the roll.
- **The runway inspection.** No document prompted either crew member to consider requesting one, so the absence of a request cannot be attributed to fatigue.

It is unlikely that the duty circumstances contributed to the occurrence. The exposure is nonetheless a standing feature of the schedule. The evening Tarlton Springs rotation routinely places the last sector in the final hour of a 12-hour duty. It does so at an aerodrome with:

- no company engineering
- no alternate within 200 nm at night
- no crewing resilience.

As a result, a single unfit crew member converts a four-sector day into a five-sector day on the morning of operation. It is likely that this exposure increases risk on future operations.

Individual Actions

No runway inspection requested before the first departure after last light

It was virtually certain that no runway inspection was requested before SB512 departed. Neither the CVR nor the tower's recorded audio contained any request for an inspection, any offer of one, or any mention of wildlife before departure. The aerodrome's daily inspection log for 18/06/2026 showed a single inspection, at 07:35 local before the first arrival, with the result 'movement area clear'. No later inspection was carried out or requested.

An inspection was available. Aerodrome Manual 6.3 provided for inspections outside the scheduled time at the request of the reporting officer or a pilot in command. The reporting officer confirmed that one would have been carried out if asked.

Both crew members stated that they did not consider requesting one. The captain had expected Tarlton Springs to be on the Appendix 8A list and was surprised to find it was not. Under Part A 8.14.3, no specific action was required at an unlisted aerodrome. Nothing in the operator's documents put an inspection request before either crew member at Tarlton Springs, and nothing in the exchange with the tower did so either. This was therefore an omission that no procedure assigned to anyone, rather than a departure from a required step. The reasons Tarlton Springs was not listed are discussed under Risk Controls.

It was likely that the absence of an inspection contributed to the animals being on the runway during the roll. An inspection would have covered the 1,050 m to 1,250 m section, where the fence pad and both carcasses were found, and aerodrome staff disperse animals when they find them.

The evidence against a higher level of confidence was also considered. An inspection carried out some minutes before the roll could not guarantee that the runway would stay clear. The first officer described the animals coming out of the dark on the left of the strip and crossing the centreline, which is consistent with animals moving onto the runway from beyond it. For that reason the influence of the omission is put at likely, not higher.

Take-off rejected 6 kt above V1

It was virtually certain that the take-off was rejected above V1. The FDR showed V1 called at 112 kt at 07:30:29. Both power levers were retarded to flight idle from 07:30:32 at 118 kt, three seconds and six knots past V1. The CVR recorded the whole decision in one exchange: the first officer's sighting call at 07:30:31 and the captain's 'Stopping. I have control.' at 07:30:32. The captain's own account, given in interview and in his air safety report, corroborated this.

Part B 4.6.4 required the take-off to be continued at or above V1, with the decision to continue made before V1 was reached. Section 4.6 contained no guidance on an obstruction seen on the runway at any speed. It also contained no amplification for a case in which continuing was itself hazardous.

The captain described his reasoning in interview and in his air safety report. In summary, he judged that rotating into two large animals would leave the aircraft airborne at night with damage that could not be assessed, over country offering no usable option. He chose the risk whose outcome he could see. He acknowledged that the decision was outside the published procedure. The CVR also showed that the captain did not see the animals himself: he had been looking at the engine page after V1, and he made the decision on the first officer's call, within about one second.

It was extremely likely that the late rejection contributed to the overrun. The scheduled accelerate-stop distance of 1,780 m assumed that the reject would begin at or before V1. The margin between ASDR and ASDA was 67 m. When the reject began, the aircraft was 1,152 m along runway 09, with 695 m of sealed surface remaining. The stopping distance achieved from 118 kt was 757 m. Beginning the stop six knots and three seconds late used up the 67 m margin and more, even before the loss of No. 2 reverse is counted. The reduced deceleration that followed the strike is a separate factor, addressed under Occurrence Events.

This decision is one link in a longer sequence, not its origin. It followed from two kangaroos being on the sealed surface at the moment of the roll. It was also shaped by rejected take-off training that addressed only aircraft system failures below V1, which is discussed under Risk Controls. Whether runway edge lighting intensity affected how early the animals could be seen could not be resolved, and is discussed under Local Conditions.

Maximum braking and No. 1 reverse applied without delay

It was virtually certain, on recorded data throughout, that the stop was carried out quickly. The power levers reached flight idle 0.6 seconds after the decision. Maximum braking was established at 07:30:34, 2 seconds after the decision, against the operator's recorded fleet mean of 3.2 seconds, which placed the response at the fast end of the fleet distribution.

Anti-skid was active on all four wheels. Brake energy reached 84% of the certified maximum without a brake fire. The captain selected reverse on No. 1 at 07:30:36, 4 seconds after the decision; No. 2 reverse was unavailable because its propeller had been damaged in the strike. The first officer made the speed callouts and the runway-end call.

The aircraft left the seal at 21 kt and stopped 62 m into the 90 m RESA. It came to rest upright and on the extended centreline, and passengers disembarked without an evacuation.

It was very likely that this response reduced the severity of the outcome. A stop begun at the fleet mean response would have carried the aircraft an estimated further 30 to 40 m, close to the far edge of the RESA. The difference between a low-speed overrun onto a prepared surface and a departure from the RESA at speed lay in those two seconds and in the braking technique used with reverse available on one side only. The rejection was outside procedure, but the stop that followed was executed faster than the fleet average.

Occurrence Events

Two kangaroos entered the runway strip and moved onto the sealed surface

The presence and position of the animals were not in doubt. Two adult kangaroos were recovered from runway 09, at 1,188 m and 1,221 m from the threshold, on the northern half of the runway. The first officer saw them crossing from the northern side towards the south and called them at 07:30:31:

> Roos - roos, roos, on the left, they're on the -

After the aircraft stopped, the first officer told the captain that the animals came out of the dark on the left of the strip and crossed the centreline. The first officer estimated the range at first sight as around 200 m.

The route by which the animals reached the runway was less certain. The site inspection on 23/06/2026 found a continuous 140 m section of the northern perimeter fence with the bottom two wires separated from the posts. There was a clear animal pad through the fence at the 1,050 m mark, on the same side as the carcasses and 138 m before the first of them. The aerodrome's wildlife sighting log also held eleven entries of kangaroos on the sealed surface of runway 09/27. Taken together, the physical evidence pointed to the pad as the most probable entry point. However, the fence had failed along its whole 140 m, so the exact crossing point was not established. For that reason, the investigation found it very likely, rather than certain, that the two kangaroos entered the runway strip through the failed northern fence and moved onto the sealed surface.

The investigation found it extremely likely that the animals' presence on the runway contributed to the occurrence. Both the rejected take-off and the strike followed directly from it. The conditions that allowed the animals onto the runway at that time are argued in other subsections:

- the deferral of the perimeter fence repair
- the departure in the hour after last light
- the absence of a runway inspection before the departure.

No. 2 propeller and right main gear struck two kangaroos

The aircraft struck the animals at 07:30:33, at 114 kt and one second after the reject began. Three independent sources of evidence agreed, and the investigation found the strike virtually certain:

- **Recorded data:** the FDR showed a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g.
- **CVR:** the CVR captured a sound consistent with a heavy impact.
- **Physical damage:** three No. 2 propeller blades were damaged beyond repairable limits, with one blade tip separated, and animal tissue was recovered from two blade roots. The right main gear doors were distorted, and the lower right fuselage skin was abraded and punctured by propeller debris.

The consequence for the stop was direct. No. 2 propeller RPM decayed from the moment of impact, and the No. 2 reverse interlock did not make. When reverse was selected at 07:30:36, it was available on No. 1 only. The captain's call at that moment reflected this:

> Reverse - number two's not coming.

The strike therefore did more than damage the aircraft. It removed one of the two reverse units almost at the start of a stop that the scheduled performance figures assumed would have symmetric reverse. The investigation found it virtually certain that the strike contributed to the overrun, through the loss of deceleration discussed below.

The investigation also considered whether a No. 2 engine power loss during the take-off roll prompted the reject. It found this very unlikely, and the hypothesis was ruled out. Both engines were within 0.1% torque of each other until the power levers were retarded, no caution, warning or exceedance was recorded, the CVR contained no engine-related call, and the No. 2 damage and RPM decay began only at the moment of impact.

Deceleration achieved below the scheduled accelerate-stop performance

The two occurrence events converged on the stop itself.

Scheduled performance. The scheduled accelerate-stop distance required was 1,780 m, against an ASDA of 1,847 m. This left a margin of 67 m, on the assumption that the stop began at or before V1 with symmetric reverse.

What happened. Neither assumption held:

- The reject was initiated at 118 kt, six knots above V1. The distance reconstruction from recorded groundspeed placed the aircraft 1,152 m along runway 09 at that point, leaving 695 m of sealed surface.
- With reverse on No. 1 only, the achieved mean deceleration was 2.6 m/s². This was approximately 15% below the value used in the certified accelerate-stop data for that weight and runway condition.
- The stopping distance actually achieved from 118 kt was 757 m.

Outcome. The aircraft left the sealed surface at 07:30:51 at approximately 21 kt. In the graded gravel, deceleration fell to 1.9 m/s², and the aircraft came to rest 62 m beyond the runway end, on the extended centreline. The recorded data, the distance reconstruction and the physical marks all agreed: continuous tyre marks from 1,160 m to the runway end, and a nose wheel rut to 62 m into the RESA. On that basis, the investigation found it virtually certain that the achieved deceleration fell below the scheduled accelerate-stop performance.

One mechanism, two contributions. The shortfall and the overrun were the same event expressed in different terms. The 62 m overrun was the product of two contributions acting together: a stop begun 6 kt beyond V1, and deceleration about 15% below what the scheduled figures assumed. The analysis indicated that both contributions were needed to carry the aircraft past the runway end, and that neither alone would have done so. The investigation found it virtually certain that the reduced deceleration contributed to the runway overrun onto the runway end safety area.

The timing of the reject decision is argued under Individual Actions. The crew's braking response, which was at the fast end of the operator's fleet distribution, is also discussed there.

Findings

From the evidence available, the following findings are made with respect to the rejected take-off and runway overrun involving Saltbush Air SB512, VH-SBQ, at Tarlton Springs Aerodrome on 18 June 2026. These findings should not be read as apportioning blame or liability to any particular organisation or individual.

Contributing factors

- It was virtually certain that Aerodrome Standard AS-139 section 10.4 required a documented wildlife plan only above a size threshold the aerodrome did not meet. It was likely that this contributed by leaving Tarlton Springs' wildlife measures without the owner and audit trail that could have passed its sighting log to operators and put the failed fence on a funded programme, although this influence was less firmly established than that of the other contributing factors, and a documented plan would not necessarily have secured funding for the repair. (Safety Issue)
- With virtual certainty, the perimeter fence repair was deferred for three budget cycles after the northern fence was recorded as failed in April 2023 with animals observed passing through, and this very likely contributed by leaving a continuous 140 m breach, with an animal pad at the 1,050 m mark, through which the kangaroos reached the runway strip. (Safety Issue)
- Two kangaroos very likely entered the runway strip through the failed section of the northern fence, although the exact crossing point was not established, and moved onto the sealed surface of runway 09, which very likely contributed by prompting both the rejected take-off and the strike.
- The take-off was rejected 6 kt above V1, which was virtually certain, and this very likely contributed to the overrun, because the stop began at 118 kt, 3 seconds after V1, and consumed the 67 m accelerate-stop margin, which assumed a stop begun at or before V1.
- Virtually certainly, the deceleration achieved was below the scheduled accelerate-stop performance at approximately 15% below the certified value, with reverse available on No. 1 only, and this virtually certainly contributed, because together with the late start of the stop it resulted in a stopping distance of 757 m against the 695 m of sealed surface remaining.
- The departure very likely took place in the hour after last light, the peak of kangaroo movement shown in the aerodrome's own sighting log, a paper register whose completeness could not be verified, and this likely contributed because, in full darkness with no moon, the animals could not be seen until they were on the lit sealed surface.
- A runway inspection was, with virtual certainty, required only at listed aerodromes, and the listing criterion was a strike, and this very likely contributed because Tarlton Springs could not be listed in Appendix 8A despite the aerodrome's log recording kangaroos on the sealed runway on 11 occasions, so the requirement to request an inspection before the first departure after last light did not apply to SB512. (Safety Issue)
- Before the first departure after last light, no runway inspection was requested, which was virtually certain. No document prompted either flight crew member to consider one, and the aerodrome would have carried one out if asked. This likely contributed, because an inspection would have covered the section of runway where the fence pad and both carcasses were found, although an inspection some minutes before the roll could not have guaranteed that the runway stayed clear.

- Very likely, the evening service was approved as a schedule change without the route risk assessment RRA-041 being reopened, despite its daylight-only restriction, and this likely contributed, because a review against a night operation could have reached the aerodrome's wildlife exposure and the Appendix 8A listing, although such a review might have addressed only the lighting and fuel matters named in the restriction. (Safety Issue)
- The operator's rejected take-off training very likely covered aircraft system failures below V1 only, with no external obstruction scenario presented in five training cycles, and this likely contributed to the decision to reject above V1 by leaving the pilot in command to consider the question for the first time within about two seconds. However, the pilot in command stated that he knew the rule and departed from it deliberately, so the evidence also supports the reading that the training gap was a fleet-wide matter rather than a cause of this rejection. (Safety Issue)
- The No. 2 propeller and right main gear virtually certainly struck two kangaroos one second after the rejection began, and this virtually certainly contributed by disabling the No. 2 reverse, which left reverse thrust on No. 1 only during a stop with a scheduled margin of 67 m.

Other factors that increased risk

- Cabin floor proximity emergency lighting was, with virtual certainty, partially unserviceable. Two of the six escape path segments aft of row 16 on the left side were dark on battery power because of a failed common ballast. The pre-flight test could not detect the defect and it lay within the 24-month functional test interval. The lighting had no influence on this occurrence, because no evacuation was commanded and the cabin lights remained on throughout. It very likely increased risk, because the same ballast is fitted across the type and the serviceability of the fleet's floor proximity lighting was unknown. (Safety Issue)
- It was virtually certain that SB512 was the fifth sector, 11 hours into duty, 40 minutes behind schedule. The recorded response times and the absence of any omission during the roll did not support an effect on this occurrence. The duty likely increased risk, because the evening Tarlton Springs rotation routinely placed the last sector in the final hour of a long duty at an aerodrome with no alternate within 200 nm at night, and a single unfit crew member could add a fifth sector on the morning of operation.

Other findings

- It was very unlikely that the take-off commenced with a tailwind above limits. The aerodrome observations and the recorded aircraft data both showed a headwind component throughout the roll.
- A No. 2 engine power loss during the take-off roll was very unlikely. Both engines were within 0.1% torque of each other with no caution or warning until the rejection, and the No. 2 propeller RPM decay began at the moment of impact.
- With virtual certainty, maximum braking and No. 1 reverse were applied without delay, with maximum braking established 2 seconds after the decision against a fleet mean of 3.2 seconds. This very likely reduced the severity of the overrun, because a stop begun at the fleet mean response was estimated to have carried the aircraft a further 30 to 40 m, close to the far edge of the RESA.

The investigation also examined whether the runway edge lighting intensity reduced the crew's ability to detect the animals, but this could not be resolved, and no finding is made. A photometric check 7 days after the occurrence found four northern edge lights between the 900 m and 1,200

m markers at 118% to 131% of nominal, in the section the animals crossed. The pilot in command described the lighting as too bright, with everything outside it black. However, the first officer, who saw the animals, described the lighting as normal and said she saw them clearly once they were on the sealed surface. The pilot in command's non-detection is explained by his attention being on the engine page after V1, as the procedure required. No measurement exists of the lighting on the night, and nothing established a link between the measured intensity and the detection of an object on the runway.

Safety issues and actions

Documented wildlife plan required only above a size threshold the aerodrome does not meet

Aerodrome Standard AS-139 section 10.4 required a documented wildlife hazard management plan only at aerodromes used by aeroplanes above 27,000 kg MTOW or serving more than 350,000 passenger movements a year. It was virtually certain that Tarlton Springs fell below both limbs. Its heaviest scheduled aeroplane was 22,900 kg, and it handled 61,400 passenger movements in 2025. The aerodrome was therefore obliged to take reasonable measures against wildlife, but not to document them. It held no plan, no hazard assessment and no dispersal programme, and was not in breach of the standard.

This matters because the threshold left an aerodrome of this size with a wildlife programme that was undocumented, unaudited and, in practice, unfunded. A documented plan would have provided an owner and an audit trail. That could have connected the aerodrome's sighting log to the operators using the aerodrome. The log recorded 41 kangaroo entries, 38 around last light and 11 on the sealed surface. A plan could also have placed the failed northern fence on a funded programme rather than an unfunded works register.

It was likely that this contributed to the occurrence. That influence was less firmly established than the other contributing factors. A documented plan is not the only means of repairing a fence or sharing a sighting log. There was also no evidence that a plan would have secured funding for a repair the council had deferred on budget grounds.

No safety action has been recorded in response to this safety issue.

Perimeter fence repair deferred for three budget cycles

The repair of the northern perimeter fence at Tarlton Springs Aerodrome was deferred for three budget cycles, although the failed section had been known to admit animals.

In April 2023, the aerodrome operator's works register (item WR-221) recorded four displaced panels and strained wire on the northern boundary between the 700 m and 1,300 m runway markers, with 'animals observed passing through'. Replacement was estimated at \$41,000. The item was deferred as unfunded at the 2023, 2024 and 2025 budget rounds. A further entry in February 2026 (WR-263) recorded the same section deteriorating further.

When the investigator inspected the site on 23 June 2026, there was a continuous 140 m section with the bottom two wires separated from the posts. A clear animal pad ran through the fence at the 1,050 m mark. This was on the same side of the runway as the positions where both carcasses were recovered, and short of them.

A fence in this condition does not exclude kangaroos. The breach was very likely the route by which the two kangaroos reached the runway strip and then moved onto the sealed surface of runway 09. The aerodrome's own register had identified the hazard more than 3 years before the occurrence. The breach remained open on the day of the inspection.

No safety action has been recorded in response to this safety issue.

Evening service approved as a schedule change, route risk assessment not reopened

The operator's current route risk assessment for Tarlton Springs, RRA-041 issue 3 (11/2022), restricted operations there to 'daylight operations only pending review of night lighting and diversion fuel policy'. Change Management Record CMR-2025-118 was approved by the Head of Flying Operations on 22/05/2025. It introduced a scheduled evening rotation from 14/07/2025, the first company operation into Tarlton Springs after last light.

It was extremely likely that this change was treated as a schedule change only. The change record found no new hazard, answered 'No' to whether a route risk assessment review was required, and did not reference the daylight-only restriction. RRA-041 was not reopened and no issue 4 exists. The Safety Department's involvement did not extend beyond receiving the approved change. The approver has since left the company and was not interviewed.

The issue is significant because the change put company aircraft into the aerodrome after last light in direct contradiction of a live restriction in the operator's own risk assessment. The change process asked whether the change was new, rather than whether the existing assessment still held.

It was likely that a review of RRA-041 against a night operation would have reached the aerodrome's wildlife exposure and the Appendix 8A listing. However, such a review might have addressed only the lighting and fuel matters named in the restriction.

No safety action has been recorded in response to this safety issue.

Runway inspection required only at listed aerodromes, and the listing criterion is a strike

The operator's Operations Manual Part A required a runway inspection before the first departure after last light only at aerodromes listed in Appendix 8A. The only way onto that list was a reported wildlife strike. It was virtually certain that this arrangement existed, and it very likely contributed to the occurrence.

Section 8.14.2 required the pilot in command to request a runway inspection before the first departure after last light at any aerodrome listed in Appendix 8A, and not to begin the take-off roll until the result was passed. Section 8.14.3 required nothing at other aerodromes. Appendix 8A listed an aerodrome only where a wildlife strike involving a company aircraft had been reported there in the preceding 24 months. Nine aerodromes were listed, and Tarlton Springs never had been. No provision in Part A:

- asked a crew to report a sighting, or
- admitted an aerodrome to the list on any other evidence.

The list was last reviewed on 3 March 2025, four months before the evening service began.

This arrangement was the operator's own defence against this kind of occurrence, and it was not in effect at Tarlton Springs. The aerodrome's sighting log recorded kangaroos on the sealed surface on 11 occasions. Those sightings could not place the aerodrome on the list, because they were not strikes and were not reported to the company. The inspection requirement itself was sound. Its trigger could be set off only by the event the control existed to prevent.

No safety action has been recorded in response to this safety issue.

Rejected take-off training covers aircraft system failures below V1 only

It was extremely likely that the operator's rejected take-off training addressed only aircraft system failures, almost all of them below V1. The four scripted scenarios in module 4 of the recurrent training syllabus TRG-SYL-07 were:

- an engine failure at 90 kt

- an engine fire at 100 kt
- a configuration warning at 85 kt
- an engine failure at V1 plus 5 kt, with the take-off continued.

No crew had been presented with an external obstruction scenario in the five training cycles on file. The current cycle alone covered 214 crew. Operations Manual Part B section 4.6 took the same approach. It listed reject items to V1 and required the take-off to be continued at or above V1. It contained no guidance on an obstruction on the runway at any speed.

This issue matters beyond this occurrence because the gap applied to the whole fleet, not to one crew. Both flight crew members independently said that every stop or go scenario they had been given was an engine event occurring below V1.

In this occurrence, the gap likely contributed to the decision to reject above V1. It left the pilot in command to consider the question for the first time within about two seconds. However, the pilot in command stated that he knew the rule and departed from it deliberately because he judged the alternative worse. On that reading, the training gap explains the circumstances of the decision without excusing it, and it remains a fleet-wide matter in its own right.

No safety action has been recorded in response to this safety issue.

Cabin floor proximity emergency lighting partially unserviceable

The cabin floor proximity emergency lighting in VH-SBQ was virtually certainly partially unserviceable. The first officer noticed sections of the aisle floor strip unlit while walking back through the cabin during the disembarkation. A functional check on 24 June 2026 found two of the six floor proximity escape path segments aft of row 16 on the left side dark on battery power with the system armed. The fault was traced to a failed common ballast.

The defect did not influence this occurrence. No evacuation was commanded, the main cabin lights remained on throughout, and the passengers disembarked by the forward airstair in 4 minutes 8 seconds without incident.

It was very likely that the defect nonetheless increased risk. The pre-flight emergency lighting test on this type illuminates the ceiling lights and exit signs but neither illuminates nor monitors the floor segments. The defect therefore could not be detected by the crew, or by a line engineer, without a dedicated functional test. The last such test had been carried out 11 months before the occurrence, within the 24-month programme interval. In an evacuation involving fire, smoke or darkness, for example over the wing at night, two dark segments in the aft cabin could have compromised passengers' ability to find the exit path. The same ballast is fitted across the type. Because the daily test cannot detect a floor segment failure and the functional test interval is 24 months, the serviceability of floor proximity lighting across the fleet was unknown.

No safety action has been recorded in response to this safety issue.

Distribution

Title	Name
Head of Flying Operations	R. Calvert
Head of Safety	M. Falkiner
Fleet Manager, Turboprop	D. Osei
Training Manager	S. Arnott
Aerodrome Reporting Officer, Tarlton Springs	K. Pryor